

Ideation Phase Submission: Circular Carbon

1. Problem Statement

SMEs and industrial facilities struggle to identify the exact sources of their carbon emissions (leak-points) and lack actionable, tailored guidance on how to transition towards a circular economy. They need a tool that translates raw process data into actionable circular interventions that demonstrate clear CO2 and cost savings.

2. Proposed Solution

Circular Carbon is an 'Industrial Emission Leak-Point Detector & Circular Alternative Recommender'. It is an intelligent recommendation engine that takes business process data as input, automatically identifies top emission sources (leak-points), and recommends specific circular economy interventions tailored to the industry. The recommendations are generated by leveraging a Retrieval-Augmented Generation (RAG) system with an embedded knowledge base of circular interventions, case studies, and recycling loops.

3. Approach

- Data Input: Users input their process data (energy source, materials used, waste streams) through a React-based interactive dashboard.

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4. Key Features

- Emission Calculation Engine: Parses process data to accurately identify and highlight major emission sources.

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5. Technology/Implementation Plan

- Frontend: React, Vite, TypeScript, TailwindCSS, Recharts (for Dashboard & visualizations).

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